Energy-Based Control of a Full-Bridge Active Power Decoupling Circuit for a 350 kW Single-Phase Active Front End

Design note

Abstract

This note develops the control design for a full-bridge active power decoupling (APD) circuit connected in parallel with the dc link of a single-phase active front end (AFE). The buffer capacitor is operated with a *bipolar* voltage at grid frequency, which gives full utilisation of the stored energy and a bounded, sinusoidal buffer current. The plant is reformulated in terms of stored energy, which removes the square-root nonlinearity of the voltage reference and yields two cascaded loops with linear plants. Reference generation, inner current-loop design with proportional-resonant (PR) compensators, outer energy loops, stability considerations and a worked design example for 350 kW on a 16.7 Hz traction network are given.

1 Scope and problem statement

A single-phase converter exchanging active power P and reactive power Q with the grid delivers an instantaneous power

$$p_g(t) = P - S\cos(2\omega_g t + \varphi), \qquad S = \sqrt{P^2 + Q^2}, \quad \varphi = \arctan\frac{Q}{P}, \tag{1}$$

where ω_g is the grid angular frequency. The dc load absorbs the constant term; the oscillating term must be buffered. Integrating it gives the energy that must be exchanged with the storage element,

$$\Delta E_{pp} = \frac{S}{\omega_g}. \tag{2}$$

Equation (2) is the governing constraint of the whole design: it is fixed by the application and cannot be reduced by any choice of topology. Only the *utilisation* of the storage element and the distribution of the stress between voltage and current are topology-dependent [1, 8].

For the case considered here ($S \approx P = 350\,\mathrm{kW}$, $f_g = 16.7\,\mathrm{Hz}$) this gives $\Delta E_{pp} = 3.34\,\mathrm{kJ}$. A purely passive dc link limiting the ripple to 5 % of $V_{dc} = 700\,\mathrm{V}$ would require approximately 136 mF, i.e. of the order of 33 kJ of installed capacitor energy. The active solution reduces the installed energy to ΔE_{pp} itself.

2 Topology

The circuit is shown in Fig. 1. A full bridge is connected across the dc link; its ac terminals feed a series branch made of a small inductor L_b and a bipolar (film) capacitor bank C_b. The bridge is the same device class as the main converter, since the buffer voltage never exceeds V_{dc}; no pre-charge circuit is required, since the buffer starts from zero voltage.

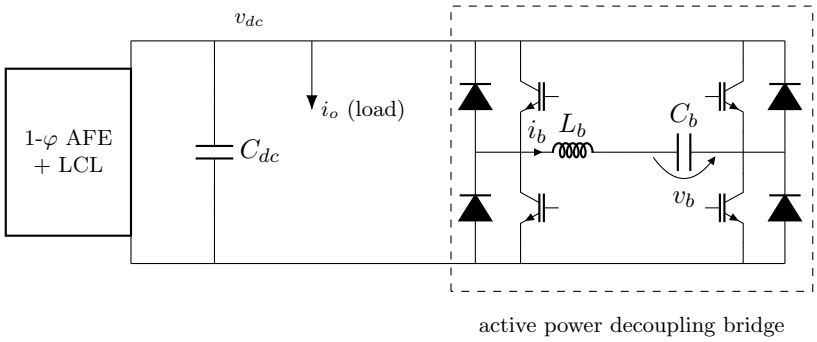


Figure 1: Full-bridge APD circuit in parallel with the dc link of the single-phase AFE.

3 Averaged model and energy formulation

Let $m \in [-1, 1]$ be the averaged modulation index of the bridge. The averaged model is

$$L_b \frac{\mathrm{d}i_b}{\mathrm{d}t} = m\, v_{dc} - v_b - R_b i_b, \tag{3}$$

$$C_b \frac{\mathrm{d}v_b}{\mathrm{d}t} = i_b, \tag{4}$$

$$C_{dc} \frac{\mathrm{d}v_{dc}}{\mathrm{d}t} = i_{in} - i_o - m\, i_b, \tag{5}$$

with R_b collecting the branch resistance (capacitor ESR, inductor and semiconductor conduction losses).

The reference trajectory for v_b is the solution of a quadratic energy balance and is therefore a square root of a sinusoid; tracking it directly gives a strongly nonlinear plant [6]. The difficulty disappears when the buffer energy is taken as the state variable:

$$W_b = \tfrac{1}{2} C_b v_b^2 + \tfrac{1}{2} L_b i_b^2. \tag{6}$$

Differentiating and substituting (3)–(4),

$$\boxed{\frac{\mathrm{d}W_b}{\mathrm{d}t} = m\, v_{dc}\, i_b - R_b i_b^2 = p_b - R_b i_b^2} \tag{7}$$

where p_b is the instantaneous power drawn by the APD from the dc link. The same transformation applied to the dc link, $W_{dc} = \tfrac{1}{2} C_{dc} v_{dc}^2$, gives

$$\frac{\mathrm{d}W_{dc}}{\mathrm{d}t} = p_g - p_o - p_b. \tag{8}$$

Both (7) and (8) are integrators driven by power. This is the basis of the "power-balance" family of APD control strategies [2, 3].

4 Steady-state reference trajectories

Impose a bipolar buffer voltage at *grid* frequency,

$$v_b^\star(t) = \hat{V}_b \cos\theta, \qquad \theta = \omega_g t + \delta, \qquad i_b^\star(t) = -\omega_g C_b \hat{V}_b \sin\theta. \tag{9}$$

Substituting into $p_b = v_b i_b + L_b i_b \dot{i}_b$,

$$p_b(t) = -\tfrac{1}{2} \omega_g C_b \hat{V}_b^2 \left(1 - \omega_g^2 L_b C_b\right) \sin 2\theta. \tag{10}$$

Matching this against the oscillating term of (1) gives two conditions:

$$\boxed{\delta = \frac{\varphi}{2} + \frac{\pi}{4}}, \qquad \boxed{C_b \hat{V}_b^2 \,(1 - k_L) = \frac{2S}{\omega_g}}, \qquad k_L \triangleq \omega_g^2 L_b C_b. \tag{11}$$

Three consequences are worth stating explicitly.

1. **Phase.** The buffer voltage is a sinusoid at ω_g, phase shifted by $\varphi/2 + \pi/4$ with respect to the grid voltage. Its product with a current in quadrature produces exactly the required $2\omega_g$ power. No absolute value, no zero-crossing singularity of the current reference: this is the structural advantage of the full bridge over the buck-type half bridge [4].

2. **Capacitor current.** $\hat{I}_b = \omega_g C_b \hat{V}_b = 2S/[\hat{V}_b(1 - k_L)]$. Since $\hat{V}_b \le V_{dc}$, the current stress is set by the dc-link voltage. Raising V_{dc} is the single most effective design lever: $C_b \propto \hat{V}_b^{-2}$ and $\hat{I}_b \propto \hat{V}_b^{-1}$.

3. **Inductor derating.** The inductor energy oscillates in anti-phase with the capacitor energy and therefore *subtracts* from the useful buffering capability by the factor $(1 - k_L)$. The branch resonance $\omega_r = 1/\sqrt{L_b C_b}$ must satisfy $\omega_r \gg \omega_g$; a practical target is $k_L \le 0.05$, i.e. $\omega_r \ge 4.5\,\omega_g$, while remaining well below the current-loop bandwidth.

The installed capacitor energy is

$$E_{C_b} = \tfrac{1}{2} C_b \hat{V}_b^2 = \frac{S}{\omega_g \,(1 - k_L)}, \tag{12}$$

which for $k_L \to 0$ equals ΔE_{pp}: the utilisation factor is unity, i.e. the theoretical minimum installed energy is reached.

The required modulation index follows from (3):

$$\hat{m} \simeq \frac{\hat{V}_b \,(1 - k_L)}{V_{dc}} \le m_{\max}, \tag{13}$$

so the inductor voltage partially cancels the capacitor voltage and slightly relaxes the modulation limit.

5 Control architecture

The controller has an inner current loop and two outer energy loops (Fig. 2).

5.1 Inner current loop

With v_b used as a feed-forward term, the plant seen by the current controller is $G_i(s) = 1/(sL_b + R_b)$ preceded by the computation and modulation delay, approximated by $e^{-1.5sT_s}$ with T_s the sampling period. Since $i_b^\star$ is sinusoidal at ω_g, a PR compensator in the stationary frame gives zero steady-state error:

$$G_c(s) = K_p + \sum_{h \in \{1,3,5\}} \frac{2K_{r,h}\,\omega_c s}{s^2 + 2\omega_c s + (h\omega_g)^2}. \tag{14}$$

The damped form is mandatory here: the traction network frequency is not perfectly constant, and an undamped resonant term would be detuned. A cut-off $\omega_c = 2\pi\,(0.5 \ldots 1)$ rad/s is a reasonable compromise; if the tolerance band is wide, make ω_g adaptive from the PLL [12]. The harmonic terms at $3\omega_g$ and $5\omega_g$ compensate dead-time distortion.

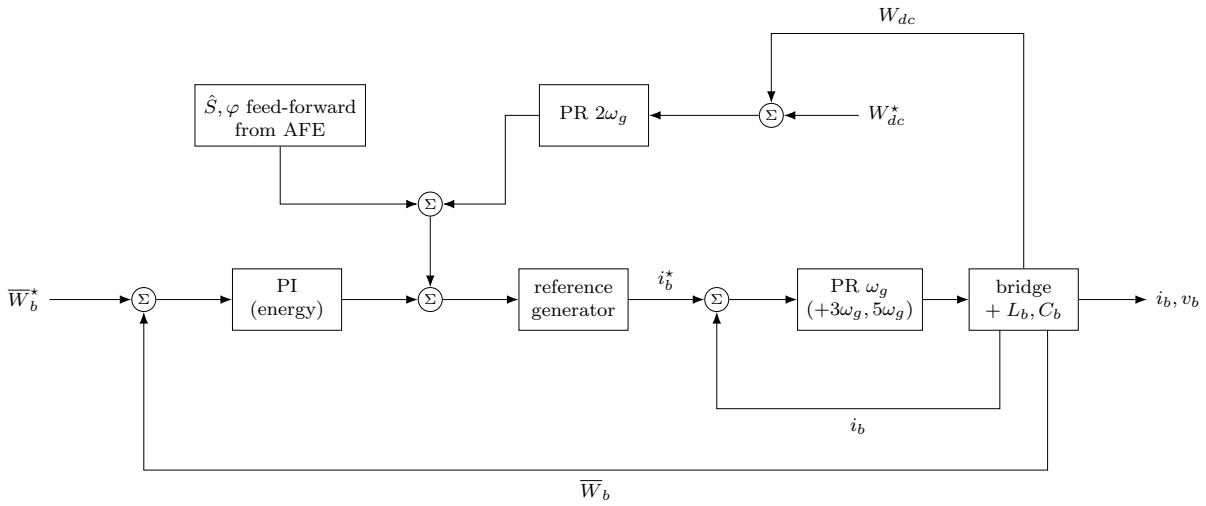


Figure 2: Control structure. Feed-forward sets the bulk of the ripple-power command; the $2\omega_g$ resonant loop on the dc-link energy removes the residual; the slow energy loop replenishes the APD losses.

Proportional gain from the desired crossover ω_{ci}:

$$K_p = \omega_{ci} L_b, \qquad \omega_{ci} \le \frac{2\pi f_{sw}}{10}, \tag{15}$$

and $K_{r,1}$ is chosen to place the loop gain at ω_g some 30 dB to 40 dB above unity; $K_{r,1} \approx (20 \dots 50) K_p$ is a usable starting point, verified on the Bode plot of $G_c(s) G_i(s) e^{-1.5 s T_s}$.

Note that $\omega_r = 1/\sqrt{L_b C_b}$ lies inside the current loop; the branch resonance is damped by the loop provided $\omega_{ci} \ge 3\omega_r$.

5.2 Buffer energy loop

The APD is lossy, so the mean value of p_b must equal the APD losses, otherwise $\overline{W}_b$ drifts. A slow PI acting on $\overline{W}_b - \overline{W}^\star_b$ corrects $\hat{V}_b$. Linearising $\overline{W}_b = \frac{1}{4} C_b \hat{V}_b^2$,

$$\frac{\partial \overline{W}_b}{\partial \hat{V}_b} = \tfrac{1}{2} C_b \hat{V}_b. \tag{16}$$

A bandwidth of $f_g/10$ is appropriate; the loop must not react within one grid period. $\overline{W}_b$ is obtained by low-pass filtering $\frac{1}{2} C_b v_b^2 + \frac{1}{2} L_b i_b^2$, or by a notch at $2\omega_g$.

5.3 DC-link energy loop

Feed-forward alone cannot cancel the ripple: errors in $\hat{S}$, in C_b (tolerance and voltage dependence of film capacitors), and in the dead-time model leave a residual $2\omega_g$ component. A PR compensator at $2\omega_g$ acting on W_{dc} (not on v_{dc}, so that the plant is the linear integrator of (8)) produces a correction to the ripple-power command. This is the "harmonic suppression" idea in the classification of [2].

At 16.7 Hz this loop is demanding. The residual dc-link ripple is

$$\Delta V_{dc,pp} \simeq \frac{P_{res}}{\omega_g V_{dc} C_{dc}}, \tag{17}$$

so with $C_{dc} = 5\,\text{mF}$ and $V_{dc} = 700\,\text{V}$ a 1 % residual ripple power (3.5 kW) already produces about 9.5 V peak-to-peak. An attenuation of at least 40 dB at $2\omega_g$ must therefore be verified on the sensitivity function, not merely assumed.

5.4 Reference generation

1. Obtain $\hat{S}$ and φ from the AFE current controller (they are already available as the amplitude and phase of the grid-current reference), and θ_g from the PLL.
2. Add the correction from the $2\omega_g$ loop to obtain the effective ripple-power command $\hat{S}^\star$.
3. Compute $\hat{V}_b = \sqrt{2\hat{S}^\star/[\omega_g C_b(1-k_L)]}$, limited to $\hat{V}_{b,\max}$, and $\delta = \varphi/2 + \pi/4$.
4. Generate $v_b^\star$, $i_b^\star$ from (9) using $\theta = \theta_g + \delta$.

Because $\hat{V}_b \propto \sqrt{\hat{S}}$, the modulation index falls at partial load and the relative dead-time distortion grows. Below roughly 30 % load the dc-link capacitor alone is usually sufficient, and the APD can be disabled; the transition must be ramped over several grid periods.

6 Stability considerations

Inner loop. With $\omega_{ci} = 2\pi \cdot 300\,\mathrm{Hz}$ and a double-update scheme at $f_s = 2f_{sw}$, the delay contributes $1.5\,\omega_{ci}T_s$ of phase lag. The phase margin follows directly from the Bode plot of $G_c(s)G_i(s)e^{-1.5sT_s}$; the resonant terms reduce the margin locally and must be included in the plot, not added afterwards.

Cascade separation. The three loops are separated by more than a decade in bandwidth ($\approx 300\,\mathrm{Hz},\ 2\omega_g,\ f_g/10$), so a sequential design is legitimate. The dc-link *voltage* loop of the main AFE must either be slower than $2\omega_g/10$ or include a notch at $2\omega_g$; otherwise it competes with the APD for the same ripple and the two controllers fight each other.

Periodic nature of the plant. The modulation index $m(t) \approx v_b(t)/v_{dc}$ varies sinusoidally at ω_g with a large amplitude, so the small-signal model is linear time-periodic. Classical LTI margins computed at one operating point are indicative only. Two practical checks: (i) evaluate the loop gain at the extremes of the trajectory ($|m| \to \hat{m}$ and $m \to 0$) and take the worst case; (ii) verify the Floquet multipliers of the periodic closed loop, or equivalently run a harmonic state-space / time-domain sweep. Lyapunov-based designs avoid the issue altogether by construction [11, 6].

DC-link interaction. Seen from the dc node, the APD behaves as a controlled power sink with a strong $2\omega_g$ component. An impedance-based (Nyquist) check of the source–load pair at the dc link, including the AFE voltage loop and the downstream inverter, is recommended before commissioning.

Saturation. If $\hat{V}_b(1-k_L)$ approaches $m_{\max}V_{dc}$ the bridge loses controllability over part of the period. The reference generator must limit $\hat{V}_b$ and the outer loops must have anti-windup; the graceful degradation is a partial decoupling with the residual ripple absorbed by C_{dc} [9].

7 Design example

$S = 350\,\mathrm{kV\,A}$, $f_g = 16.7\,\mathrm{Hz}$ ($\omega_g = 104.9\,\mathrm{rad/s}$), $V_{dc} = 700\,\mathrm{V}$, $\hat{V}_b = 600\,\mathrm{V}$, $L_b = 200\,\mu\mathrm{H}$, $f_{sw} = 3\,\mathrm{kHz}$ with unipolar (three-level) PWM.

Solving (11) for C_b gives the values of Table 1. The last column repeats the exercise for a 1100 V dc link, which is still within the same 1700 V device class.

Table 1: Design example.

Quantity	Symbol	$V_{dc} = 700\,\mathrm{V}$	$V_{dc} = 1100\,\mathrm{V}$
Buffer voltage amplitude	$\hat{V}_b$	600 V	950 V
Buffer capacitance	C_b	19.4 mF	7.5 mF
Inductor factor	k_L	0.043	0.017
Branch resonance	f_r	81 Hz	130 Hz
Peak buffer current	$\hat{I}_b$	1218 A	750 A
RMS buffer current	$I_{b,rms}$	861 A	530 A
Installed capacitor energy	E_{C_b}	3.48 kJ	3.39 kJ
Peak modulation index	$\hat{m}$	0.82	0.85
Current ripple, pk-pk	ΔI_{pp}	146 A	229 A
Bridge apparent power	$\hat{V}_b\hat{I}_b/2$	365 kV A	356 kV A

Both columns use $L_b = 200\,\mu\mathrm{H}$ and $f_{sw} = 3\,\mathrm{kHz}$. For the 1100 V case the current ripple is 30 % of $\hat{I}_b$; raising L_b to 400 µH brings it to 115 A at the cost of $k_L = 0.034$ and $f_r = 91\,\mathrm{Hz}$.

Controller parameters for the 700 V case: $\omega_{ci} = 2\pi \cdot 300\,\mathrm{rad/s}$, $K_p = \omega_{ci} L_b = 0.377\,\Omega$, $K_{r,1} = 30K_p$, $\omega_c = 2\pi \cdot 1\,\mathrm{rad/s}$; buffer energy loop at 1.7 Hz; dc-link resonant loop at $2\omega_g = 33.3\,\mathrm{Hz}$ with $\geq 40\,\mathrm{dB}$ of loop gain at that frequency.

Two observations on the numbers. First, the bridge apparent power is essentially equal to S: an APD in parallel processes the full pulsating power, and its silicon is comparable to that of the main bridge. Second, the buffer rms current makes the capacitor ESR a first-order design constraint: at 861 A rms, an equivalent series resistance of 1 mΩ already dissipates about 740 W in the bank alone.

8 Practical notes

- **Start-up.** v_b starts from zero; ramp $\hat{V}_b$ over ten or more grid periods (0.6 s at 16.7 Hz). No pre-charge circuit and no minimum-voltage constraint, unlike a boost-type buffer.
- **Dead time.** v_b crosses zero when i_b is at its peak, so the current polarity is unambiguous where the modulation index is small: the dead-time error appears as an almost constant voltage offset and is straightforward to compensate. The residual shows up at $3\omega_g$ and $5\omega_g$ and is handled by the corresponding resonant terms in (14).
- **Capacitor technology.** The bank must be non-polarised: metallised polypropylene film. Specify on rms current and $\mathrm{d}v/\mathrm{d}t$, and account for the capacitance drift with temperature and ageing, which the $2\omega_g$ loop must absorb.
- **Interleaving.** At 1218 A peak, two interleaved full-bridge units halve the device current and double the effective ripple frequency. This is usually cheaper than a single very large module and adds partial redundancy.
- **Fault handling.** On loss of the APD the dc-link ripple returns to the passive value. The supervisory logic should derate the transferred power rather than trip, since $\Delta V_{dc} \propto P$.
- **Partial decoupling.** Allowing 10 % to 15 % ripple on C_{dc} and sizing the APD only for the remainder reduces its rating roughly in proportion [9].

9 Alternative control formulations

- **Virtual impedance / resonator emulation.** The APD is controlled so that it emulates an LC resonator tuned at $2\omega_g$ seen from the dc link [7]. No explicit estimate of $\hat{S}$ is needed, which makes it attractive when the turbine power varies quickly; the trade-off is a slower transient response.
- **Nonlinear and Lyapunov-based control.** Direct treatment of the nonlinear voltage reference with guaranteed convergence [6, 11]; also the natural route to a passivity-based (IDA-PBC) formulation, for which (7) is already the required energy function.
- **Predictive control.** The energy formulation provides a one-dimensional cost function (W_b tracking) that suits finite-control-set MPC, at the cost of a variable switching frequency, which is a drawback for the EMC qualification of a traction-connected converter.
- **Series voltage compensator.** A different architecture altogether: the compensator is inserted in series with the dc link and only synthesises the ripple voltage, which lowers its VA rating at the price of carrying the full dc current [10].

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